7 A washing machine has an electric motor and an electric heater. Fig. 7.1 shows a simplified circuit diagram for the washing machine.

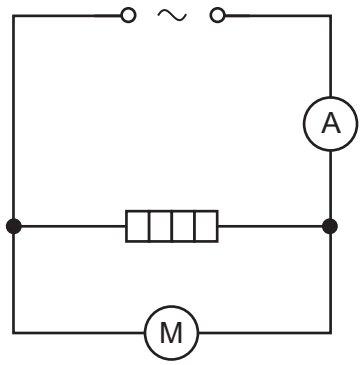

Fig. 7.1

The heater has a resistance of 25 Ω and the power supply has an electromotive force (e.m.f.) of 230 V.

(a) State the meaning of electromotive force.

.....

..... [2]

(b) State the potential difference (p.d.) across the heater.

p.d. = ... [1]

(c) Calculate the current in the heater.

Current = ... [2]

(d) The current in the motor is 1.6 A.
Determine the reading on the ammeter in Fig. 7.1. Explain your answer.

Ammeter reading .....

Explanation .....

.....

[2]

[Total: 7]

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